CPO: Cluster-Referenced Preference Optimization

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1 Introduction and Intuition

Preference-based fine-tuning of language models has two prominent paradigms:

- **DPO (Direct Preference Optimization):** relies on *pairs* of outputs (y_w, y_ℓ) for the same prompt, and optimizes a logistic loss on their log-probability difference. The reference point for each sample is the *other response in the pair*.

- **KTO (Kahneman–Tversky Optimization):** relies on *single labeled outputs*, and evaluates each outcome relative to a *global reference point*, given by the average KL divergence of the policy from the reference model. This mirrors Kahneman–Tversky's prospect theory: outcomes are judged as gains or losses relative to a baseline status quo.

These are two extremes: DPO is fully local (pairwise), KTO is fully global (single baseline). Human judgments, however, are often contextual: outcomes are judged relative to the norms of a *reference group* or *cluster*, not only the global baseline or a single counterexample. For example, medical answers may be evaluated differently depending on the affinity group of annotators, or stylistically similar outputs may be compared against each other.

We therefore propose a generalization: *Cluster-Referenced Preference Optimization (CPO)*. Each example is evaluated relative to the reference point of its cluster. This framework unifies:

- KTO (all data in one cluster $\Rightarrow$ single reference point),

- DPO (each cluster of size two containing a win/loss pair $\Rightarrow$ pairwise logistic).

2 Setup and Notation

We are given singly-labeled data

$$\mathcal{D} = \{(x_i, y_i, s_i)\}_{i=1}^n, \qquad s_i \in \{\mathrm{D}, \mathrm{U}\} \text{ (desirable / undesirable).}$$

Let $\pi_\theta(y|x)$ denote the trainable policy, and $\pi_{\mathrm{ref}}(y|x)$ the frozen reference model.

Define the log-ratio reward proxy

$$r_i := \log \frac{\pi_\theta(y_i|x_i)}{\pi_{\mathrm{ref}}(y_i|x_i)}.$$

The dataset is partitioned (hard or soft) into K clusters $C_1, \ldots, C_K$. Each example i belongs to cluster $c(i)$, or has membership weights w_{ik}.

3 Cluster Reference Points

For each cluster k, define a baseline "cost of deviation" as

$$z_k \;=\; \mathbb{E}_{x \in C_k}\left[\, \mathrm{KL}\big(\pi_\theta(\cdot|x) \,\|\, \pi_{\mathrm{ref}}(\cdot|x)\big)\,\right].$$

Each example i uses the cluster reference $z_{c(i)}$. (For soft clusters, use $z_i = \sum_k w_{ik} z_k$.)

Practical estimation. Maintain z_k as an exponential moving average (EMA) of minibatch KL estimates:

$$z_k \leftarrow (1-\rho)\, z_k + \rho\, \widehat{\mathrm{KL}}_{\mathrm{mb},k}, \quad \text{with } \widehat{\mathrm{KL}}_{\mathrm{mb},k} = \frac{1}{|B_k|} \sum_{x \in B_k} \mathrm{KL}\big(\pi_\theta(\cdot|x) \,\|\, \pi_{\mathrm{ref}}(\cdot|x)\big).$$

4 Cluster-Prospect Loss

For each sample i, define the signed margin to its cluster reference:

$$m_i := \begin{cases} r_i - z_{c(i)} & \text{if } s_i = \mathrm{D}, \\ z_{c(i)} - r_i & \text{if } s_i = \mathrm{U}. \end{cases}$$

With logistic σ, slope $\beta > 0$, and asymmetry weights (λ_D, λ_U), define the prospect value

$$v_i = \begin{cases} \lambda_D\, \sigma(\beta m_i) & \text{if } s_i = \mathrm{D}, \\ \lambda_U\, \sigma(\beta m_i) & \text{if } s_i = \mathrm{U}. \end{cases}$$

The unary (KTO-like) loss is

$$\mathcal{L}_{\mathrm{unary}} \;=\; \mathbb{E}_i \left[\lambda_{s_i} - v_i\right].$$

5 In-Cluster Contrastive Term

Optionally, for pairs (i, j) of desirable/undesirable examples within the *same prompt and cluster*, add a DPO-style term:

$$\mathcal{L}_{\text{pair}} \;=\; \mathbb{E}_{(i,j):\, c(i)=c(j),\, s_i=\text{D},\, s_j=\text{U}}\Big[-\log \sigma\big(\beta(r_i - r_j)\big)\Big].$$

6 Unified Objective

$$\mathcal{L} = (1-\alpha)\,\mathcal{L}_{\text{unary}} + \alpha\,\mathcal{L}_{\text{pair}} + \eta\,\mathbb{E}_x\left[\,\text{KL}\big(\pi_\theta(\cdot|x)\,\|\,\pi_{\text{ref}}(\cdot|x)\big)\,\right],$$

with mixing parameter $\alpha \in [0, 1]$ and KL regularization coefficient $\eta \geq 0$. It is optional but stabilizing.

7 Recovering Extremes

- **KTO:** If $K = 1$ and all $w_{i1} = 1$, the unary term equals the KTO/HALO loss (exact if using $r = \log \pi_\theta - \log \pi_{\text{ref}}$ or $\beta = 1$ under $r^{(\beta)}$). Setting $\alpha = 0$ recovers exact KTO; for $\alpha > 0$ the objective is KTO plus a uniform pairwise auxiliary.

- **DPO:** If clusters are size-two win/loss pairs (one-hot memberships) and $\alpha = 1$, then $\mathcal{L}$ reduces to canonical DPO (plus optional KL).

Proof. **Case (1).** We will prove that under the assumptions $K = 1$ and $\alpha = 0$, the CPO objective reduces to the KTO objective.
Assumption:

1. $K = 1$: There is only one cluster, C_1, containing the entire dataset D.

2. $\alpha = 0$: The pairwise loss term is completely turned off.

Derivation: Substitute $\alpha = 0$ into the $\mathcal{L}_{\text{CPO}}$ objective:

$$\mathcal{L}_{\text{CPO}} = \mathcal{L}_{\text{unary}} + \eta\,\mathbb{E}_x\left[\,\text{KL}\big(\pi_\theta(\cdot|x)\,\|\,\pi_{\text{ref}}(\cdot|x)\big)\,\right],$$

$\mathcal{L}_{\text{unary}}$ under the assumption $K = 1$, since there is only one cluster ($C_1 = D$), the cluster ID for every sample i is the same: $c(i) = 0$. Consequently, there is only one reference point, z_0. This reference point is calculated over the entire dataset, making it a **global reference point**. Regardless of whether z_0 is defined as the average KL divergence or the average reward r_i, it is an expectation taken over the entire dataset.

Let's examine the margin m_i for any sample i in the dataset: The cluster reference $z_{c(i)}$ is always z_0.

$$m_i := \begin{cases} r_i - z_0 & \text{if } s_i = \mathrm{D}, \\ z_0 - r_i & \text{if } s_i = \mathrm{U}. \end{cases}$$

The unary loss $\mathcal{L}_{\text{unary}}$ therefore becomes:

$$\mathcal{L}_{\text{unary}} \;=\; \mathbb{E}_i \left[\lambda_{s_i} - \lambda_{s_i} \sigma(\beta m_i) \right].$$

where m_i is computed relative to the single global reference z_0.

Case (2). We will prove that under the assumptions that clusters are size-two win/loss pairs and $\alpha = 1$, the CPO objective reduces to the DPO objective.

Assumption:

1. $\alpha = 1$: The unary loss term is completely turned off.

2. **Clustering Scheme:** Each cluster C_k corresponds to a single prompt x_k and contains exactly two samples: one desirable (winning) response y_w and one undesirable (losing) response y_l. Let's denote the sample for the winning response as w and the losing response as l. Thus, $C_k = \{w, l\}$ where $s_w = D$ and $s_l = U$.

Derivation: Substitute $\alpha = 1$ into the $\mathcal{L}_{\text{CPO}}$ objective:

$$\mathcal{L}_{\text{CPO}} = \mathcal{L}_{\text{pair}} + \eta \, \mathbb{E}_x \left[\mathrm{KL}\big(\pi_\theta(\cdot|x) \,\|\, \pi_{\text{ref}}(\cdot|x)\big) \right],$$

The definition of $\mathcal{L}_{\text{pair}}$ is:

$$\mathcal{L}_{\text{pair}} \;=\; \mathbb{E}_{(i,j):\, c(i)=c(j),\, s_i=\mathrm{D},\, s_j=\mathrm{U}} \Big[-\log \sigma\big(\beta(r_i - r_j)\big) \Big].$$

The expectation is over all pairs of samples (i, j) that are in the same cluster with sample i being desirable and sample j undesirable. According to our clustering assumption, every cluster C_k is by definition a pair (w, l) that satisfies these exact conditions. The set of all such pairs across all clusters is therefore equivalent to the set of all (winner, loser) pairs (y_w, y_l) from the standard preference dataset. Therefore, the expectation over in-cluster pairs becomes an expectation over the dataset. For each pair, i is the winner w and j is the loser l. Substituting these into the expression for $\mathcal{L}_{\text{pair}}$:

$$\mathcal{L}_{\text{pair}} \;=\; \mathbb{E}_{(x, y_w, y_l)} \Big[-\log \sigma\big(\beta(r_w - r_l)\big) \Big].$$

□

8 Lemma: Soft-Cluster Interpolation Between KTO and DPO

Lemma. Let examples have soft memberships $w_{ik} \geq 0$ with $\sum_k w_{ik} = 1$, and define per-example references $z_i = \sum_k w_{ik} z_k$. Suppose:

1. If $K = 1$ and all $w_{i1} = 1$, then the unary term equals the KTO loss. Setting $\alpha = 0$ recovers exact KTO.

2. If each prompt forms a disjoint two-element cluster with one desirable and one undesirable, and memberships are one-hot, then with $\alpha = 1$ the CPO loss reduces to canonical DPO (plus optional KL).

Moreover, the objective $\mathcal{L}(\theta; w)$ is continuous in the weights w_{ik} and in α. Thus CPO interpolates continuously between KTO and DPO as w and α vary.

Proof sketch. Case (1) reduces to KTO by collapsing all examples to one cluster. Case (2) reduces to DPO because only win–loss pairs with co-membership 1 contribute to the pairwise term. Continuity follows since both the unary and pairwise terms depend on w only through linear combinations $z_i = \sum_k w_{ik} z_k$ and $\sum_k w_{ik} w_{jk}$. $\square$

Let $f(w, \alpha) = (1 - \alpha) L_{\text{unary}}(w) + \alpha L_{\text{pair}}(w) + \eta L_{\text{KL}}(\theta)$. For every fixed θ (treat KL term as independent of w in this argument or continuous in w), each component L_{unary} and L_{pair} is built from finite sums, products and sigmoid evaluations of linear forms in w. All these operations are continuous; finite linear combinations of continuous functions are continuous; multiplication by continuous α keeps continuity.

9 Training Procedure (Hard Clusters; Minibatch + EMA)

Algorithm 1 CPO Training Loop with EMA Cluster References (hard clusters)

Inputs: Dataset $\mathcal{D}$ with labels $s_i \in \{\mathrm{D}, \mathrm{U}\}$ and cluster IDs $c(i) \in \{1, \ldots, K\}$; policy π_θ; reference π_{ref}; hyperparams $\alpha, \beta, \lambda_D, \lambda_U, \eta, \rho$.
Init: For $k = 1, \ldots, K$, set $z_k \leftarrow 0$ (or estimate from a pilot batch).
for each training step **do**
    Sample a minibatch $B \subset \mathcal{D}$.
    for each cluster k with at least one sample in B **do**
        $B_k \leftarrow \{i \in B : c(i) = k\}$.
        $\widehat{\mathrm{KL}}_{\mathrm{mb},k} \leftarrow \frac{1}{|B_k|} \sum_{i \in B_k} \mathrm{KL}\big(\pi_\theta(\cdot|x_i) \,\|\, \pi_{\text{ref}}(\cdot|x_i)\big)$.
        $z_k \leftarrow (1 - \rho)\, z_k + \rho\, \widehat{\mathrm{KL}}_{\mathrm{mb},k}$ ▷ EMA update (stop-grad).
    end for
    Compute $r_i = \log \pi_\theta(y_i|x_i) - \log \pi_{\text{ref}}(y_i|x_i)$ for $i \in B$.
    Compute margins m_i and unary loss $\mathcal{L}_{\text{unary}}$.
    if $\alpha > 0$ **then**
        Form in-cluster pairs (same prompt) and compute $\mathcal{L}_{\text{pair}}$.
    else
        $\mathcal{L}_{\text{pair}} \leftarrow 0$.
    end if
    Compute optional global KL penalty $\mathcal{L}_{\mathrm{KL}}$.
    Total loss: $\mathcal{L} \leftarrow (1 - \alpha)\mathcal{L}_{\text{unary}} + \alpha \mathcal{L}_{\text{pair}} + \eta \mathcal{L}_{\mathrm{KL}}$.
    Backpropagate and update θ.
end for

10 Reference Estimation Variants (Practical)

Several variants exist for estimating z_k:

1. **KL-baseline (principled):** $z_k \approx \mathbb{E}_{x \in C_k} \mathrm{KL}(\pi_\theta(\cdot|x) \,\|\, \pi_{\text{ref}}(\cdot|x))$.

2. **Log-ratio mean (cheap proxy):** $z_k \approx \mathbb{E}_{(x,y) \in C_k}[r_i]$ or mean of undesirable examples only $z_k \approx \mathbb{E}_{(x,y,s=U) \in C_k}[r_i]$ to make "losses" anchor the reference.

3. **Robust baselines:** Use median or Huber-smoothed averages of r_i to resist outliers.

4. **Soft clusters:** For soft memberships, compute $z_i = \sum_k w_{ik} z_k$; optionally regularize w_{ik} with entropy.

11 Hyperparameters and Knobs

- K: number of clusters ($K = 1$ yields KTO, K large with $\alpha = 1$ yields DPO).
- α: dial between unary (KTO-like) and pairwise (DPO-like) pressure.
- β: slope parameter controlling saturation; higher β makes the updates more "winner-take-all".
- λ_D, λ_U: asymmetry weights (loss aversion).
- η: global KL regularization weight.
- ρ: EMA update rate for z_k.

12 Connections and Notes

- **Unifying principle:** CPO is a reference-anchored unary objective with optional in-cluster contrasts. It recovers KTO (single cluster, $\alpha = 0$) and DPO (pairwise clusters, $\alpha = 1$).
- **Robustness:** Cluster references adapt to heterogeneous annotator norms, unlike global KTO, and CPO does not require complete preference pairs, unlike DPO.
- **Gradient behavior:** The unary and pairwise terms produce aligned logistic gradients, differing mainly in how reference points are defined.
- **Continuity:** The soft-cluster formulation provides a continuous path between global and local reference anchoring, interpolating KTO and DPO.

13 Discussion

CPO provides a continuum between pairwise and global-reference optimization:

- It adapts to heterogeneous annotator norms by anchoring to cluster-specific references.
- It preserves the efficiency of single-label data (as in KTO).
- It retains the contrastive sharpness of DPO when clusters collapse to pairs.

14 Remarks

- **Robustness to Annotator Heterogeneity** : A single global reference point (as in KTO) can be skewed by noisy or inconsistent subgroups of annotators. By creating cluster-specific references, CPO has the potential to be more robust. For instance, if one cluster of annotators is consistently more lenient, their "desirable" ratings will be anchored against their own baseline, not pulled down by a stricter global average.
- **Unifying principle:** CPO provides a much richer mental model for understanding preference optimization and moves the field beyond a dichotomy between pairwise and unary methods.

15 Open Questions

The entire CPO framework relies on the existence of meaningful clusters C_k.

1. **Pre-defined vs. Learned Clusters**: Are clusters assumed to be pre-defined based on metadata (e.g., annotator ID, prompt source, topic label)? Or are they to be learned from the data itself?
2. **If Learned, How?** If clusters are learned, what features should be used? Prompt embeddings? Response embeddings (from the policy model or a separate encoder)? A combination? Should clustering be a static preprocessing step, or should the clusters be updated dynamically during training (i.e., online clustering)?
3. **Sensitivity to Cluster Quality:** Poorly formed clusters could lead to noisy and unstable reference points z_k, potentially performing worse than having a single global reference.
   (a) **Soft Clustering (Gaussian Mixture Models)**
   The problem: A sample lying just across a boundary from a

nearly identical sample will be assigned a completely different reference point, which can be suboptimal.
The solution: Use a soft clustering method(GMM can be a good candidate). A sample near the boundary of two clusters will have its reference point be an interpolation of the two cluster references (e.g., $z_i \approx 0.5 * z_1 + 0.5 * z_2$). This creates a much smoother loss landscape and makes the system far less sensitive to the exact placement of cluster boundaries.

(b) **Treating α as a "Confidence Dial"**.
The Problem: We may have a mix of high-confidence clusters (e.g., from clean metadata like annotator ID) and low-confidence clusters (e.g., from noisy learned embeddings). A single setting for the whole model might not be right.
The solution: The α parameter in $\mathcal{L}$ is the perfect tool for this. For high confidence in clusters we should set α to a low value (e.g., 0.1 to 0.3) to maximize the benefit of the contextual reference points. For low confidence in clusters, we should set α to a higher value (e.g., 0.5 to 0.8). This reduces the reliance on the potentially unstable z_k values and leans more heavily on the robust, local DPO-style signal. In the extreme, $\alpha = 1$ ignores clusters entirely.
Practical point: Treat α as a hyperparameter that reflects our trust in our clustering methodology. It is our primary lever for managing the trade-off between contextual adaptation and robustness.

4. The In-Cluster Contrastive Term ($\mathcal{L}_{\text{pair}}$): The formulation is for pairs (i, j) with $s_i = D, s_j = U$ within the same prompt and cluster. Handling Multiple Examples: What if a prompt has multiple "desirable" and multiple "undesirable" responses within the same cluster? Does this term create a loss for every possible (D, U) pair? Clarifying the exact sampling or pairing strategy is important.

   (a) Strategy 1: Random 1-to-1 Pairing

   (b) Strategy 2: Hard-Negative / Hard-Positive Mining

5. **EMA Dynamics:** What happens if a cluster is sparsely represented in minibatches, leading to a stale z_k estimate?

6. **Interactions:** As K increases and clusters become smaller, does the optimal α shift towards 1 (pairwise)?

7. **EMA Dynamics 2:** The definition of z_k implies that z_k is a function of the current policy parameters θ. Therefore, a true gradient descent on the CPO loss should include terms from $\partial z_k/\partial\theta$. The loss function $\mathcal{L}$ depends on $z_k(\theta)$. By applying stop-grad, the algorithm treats z_k as a fixed constant during the backpropagation step for a given mini-batch. This means the algorithm is optimizing a surrogate objective where the reference points are considered fixed, rather than the true objective where the reference points co-evolve with the policy. This simplification can lead to training instabilities or prevent convergence to a true optimum of the theoretical loss.

   - The use of stop-grad on the z_k update means the practical algorithm does not take the true gradient of the theoretical loss $\mathcal{L}(\theta, z_k(\theta))$. This makes the optimization a heuristic without a clear objective.
   - **Proposed Solution: Re-frame the Optimization Process:** Instead of viewing it as a single joint optimization, formalize the training loop as an **alternating optimization** or **Expectation-Maximization (EM)-like** procedure. This provides a rigorous justification for treating z_k as a constant during the policy update.

     (a) (E-Step) Reference Estimation: For a fixed policy θ_t, update the estimate of the cluster reference points. This is exactly what the EMA update does.

     $$z_k^{(t+1)} \leftarrow (1-\rho)z_k^{(t)} + \rho\,\widehat{\mathrm{KL}}_{\mathrm{mb},k,\theta_t}$$

     (b) (M-Step) Policy Optimization: For fixed reference points $z_k^{(t+1)}$, update the policy parameters by taking a gradient step to minimize the CPO loss.

     $$\theta_{t+1} \leftarrow \theta_t - \gamma * \nabla_\theta \mathcal{L}(\theta_t; z_k^{(t+1)})$$

     This is precisely what stop-grad achieves: the gradient calculation for θ treats z_k as a pre-computed constant.
   - Benefits:

     (a) Theoretical Grounding: It legitimizes the use of stop-grad not as a hack, but as a necessary component of an alternating optimization scheme.

(b) Clarity : It clarifies that the algorithm is not finding a stationary point of a single objective, but is rather iteratively improving the policy and the reference points in a coupled manner. This often leads to stable and effective training in practice.

(c) No Change to Implementation: Requires no change to the code in Algorithm 1. It is purely a change in the theoretical framing that makes the algorithm description rigorous and defensible.